How Much of Context Extension Is Positional? Learning RoPE Parameters on a Frozen Language Model

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Abstract

Language models that use rotary position embeddings (RoPE) fail beyond their pretraining length. Extending them usually means fine-tuning all weights or applying a fixed rule such as YaRN. We ask how much of context extension is positional: can a *frozen* model be extended by learning only how it encodes position? We extend OLMo-2-1B from 4k to 16k tokens by training per-band frequency scales, at model, layer or head granularity, together with a per-head attention temperature. This is at most $\sim$17k parameters, about 0.001% of the model. Against grid-tuned YaRN, our best configuration (per head, started from YaRN) lowers 16k perplexity on PG-19 from 21.4 to 15.1 and more than doubles RULER needle-in-a-haystack accuracy at 16k (24.0 $\rightarrow$ 53.1), while staying within one point of the original model's 4k perplexity.

1 Introduction

Nearly every open-weight language model encodes position with rotary embeddings (RoPE; Su et al., 2024), and nearly every one degrades sharply once its input exceeds the length L it was pretrained on (Chen et al., 2023; Peng et al., 2024). OLMo-2-1B (Team OLMo, 2024), for example, is trained with $L = 4096$. Given a 16k-token book, its perplexity explodes, and it fails every RULER retrieval task (Hsieh et al., 2024) at 8k and 16k, although the whole input fits in memory (Table 1). Pretraining at longer lengths is costly, since attention compute grows quadratically with length (16 times more at 16k than at 4k), so cheap extension of existing models is a practical, widely shared problem.

Why it is hard. RoPE divides each head's query and key vectors into $d/2$ two-dimensional *bands* and rotates band j of the token at position m by angle $m\theta_j$, where $\theta_j = b^{-2j/d}$. Only the difference in angle survives the dot product, so the attention logit depends on relative distance alone:

$$s_{mn} = \sum_j \mathrm{Re}\big[\, q_{m,j}\, \overline{k_{n,j}}\, e^{\,i(m-n)\theta_j}\big]. \qquad (1)$$

Fast bands cycle many times within L and encode local order. *Slow bands*, whose wavelength $2\pi/\theta_j$ exceeds L, never complete a rotation in training; beyond L they take angles the model has never seen. A second problem remains even when the angles are fixed: with four times as many keys, each softmax spreads its mass more thinly, so attention becomes flatter than the model was trained on.

How obvious solutions fail. Position Interpolation (PI; Chen et al., 2023) divides every position by L'/L, which keeps slow-band angles in range but also compresses the fast bands; it harms the model even at 4k. YaRN (Peng et al., 2024) interpolates only the slow bands and sharpens attention with a temperature, but it applies one hand-designed rule to every head, and its default settings fall far short of settings found by a grid search on validation data (Table 1).

Task. We treat context extension as a parameter-efficient learning problem. Given a pretrained model with frozen *backbone weights* W (all of its original parameters) and native length L, we seek a small set of *positional parameters* ϕ, $|\phi| \ll |W|$, that change only how positions are encoded and let the model operate at $L' = 4L$:

$$\min_{\phi}\ \mathbb{E}_{x\sim\mathcal{D}_{L'}}\big[-\log p_{W,\phi}(x)\big], \qquad (2)$$

while preserving its behavior up to L. We learn a scale for each RoPE frequency and a per-head temperature (§2.1), starting from exactly the original model. Beyond the practical goal, this asks a scientific question: *how much of context extension is positional?* If training ϕ alone closes most of the gap to fine-tuning W, the length limit lies in how positions are encoded, not in the backbone weights.

	PPL ↓			PK ↑	RULER ↑		
Method	4k	8k	16k	16k	4k	8k	16k
Original	**16.4**	124.6	430.3	0	**72.8**	0.0	0.0
PI (×4)	109.1	118.5	110.9	0	0.0	0.0	0.0
YaRN, default	27.1	40.1	41.7	1	8.3	6.6	1.2
YaRN, tuned	19.9	22.9	21.4	80	45.7	35.4	24.0
Frequencies only, from a=0							
Shared	21.8	23.9	22.5	0	12.2	4.7	0.2
Per layer	20.0	21.3	19.5	0	39.3	21.9	1.3
Per head	20.7	21.8	19.7	0	5.8	0.8	0.0
Freq. + temperature, from a=0							
Shared	18.1	18.7	16.2	**81**	61.0	49.1	36.8
Per layer	17.9	18.4	16.0	1	59.9	43.7	33.3
Per head	18.7	19.4	17.2	0	24.5	21.3	5.9
Freq. + temperature, from YaRN							
Shared	17.3	17.8	15.4	80	61.4	49.0	36.2
Per layer	17.2	17.6	15.3	4	58.6	50.1	35.6
Per head	17.0	**17.4**	**15.1**	0	69.8	**62.9**	**53.1**
PoSE (4k text) + freq. + temp., from YaRN							
Shared	17.1	17.7	17.6	0	72.1	56.7	0.1
Per layer	17.3	17.8	18.6	0	65.8	47.5	0.0
Per head	17.1	17.6	16.5	0	59.1	43.1	1.1

Table 1: Test results. Trainable parameters: 0 for the baselines; frequency scales 64 (shared), 1,024 (per layer) or 16,384 (per head), plus 512 for the temperature. PPL: PG-19 perplexity. PK: passkey accuracy (%) at 16k with the standard prompt, 250 prompts. RULER: average over the 8 tasks (%), 500 samples each. PoSE from a=0 behaves the same as PoSE from YaRN. Bold: best.

Research questions and progress. RQ1 asks what learning RoPE parameters per model, layer or head on a frozen backbone adds beyond tuned YaRN; RQ2 whether they can be learned from 4k sequences alone with PoSE position skipping (Zhu et al., 2024); RQ3 whether slow bands should instead learn a content-dependent coordinate, as in DaRoPE (Lévy et al., 2026); and RQ4 which backbone weights must change to close what remains.

2 Experiments

2.1 Methods

Model. We extend OLMo-2-0425-1B (Team OLMo, 2024): 16 layers of 16 heads (no grouped-query attention), head dimension 128 (64 bands), RoPE base $b = 5 \times 10^5$, $L = 4096$.

Implementation. We patch the Hugging Face OLMo-2 code (Wolf et al., 2020) so that each attention layer rotates its queries and keys with its own parameters. All of ϕ sits in one float32 module (PI and YaRN are fixed settings of it), and every learned mode starts with logits identical to the stock model (unit-tested).

Learned frequencies. Following LeRoPE (Karypis et al., 2026) and AdaRoPE (Wang et al., 2026), each band's frequency gets a learned log-scale, $\hat{\theta}_{g,j} = e^{a_{g,j}}\theta_j$, where g sets the granularity: one vector for the model (*shared*, as in LeRoPE; 64 parameters), one per layer (1,024), or one per head (16,384, as in AdaRoPE), started at $a = 0$ (stock RoPE) or at YaRN's frequencies.

Learned temperature. Against flatter attention, each head multiplies its queries, and so its logits, by AdaRoPE's length-dependent scale

$$\lambda_h(i) = \frac{1}{\tau_h}\left[\ln\left(1 + \frac{\max(i+1,\, L)}{L}\right)\right]^{\gamma_h}, \quad (3)$$

with τ_h, γ_h learned per head (512 parameters) and $i + 1$ the number of keys query i sees. With τ_h alone, the temperature is constant: sharpening that helps at 16k changes the model equally at 4k. The bracket is constant up to L and grows beyond it, so γ_h sets how much a head sharpens on long contexts, separately from τ_h. Initialization: $\tau_h = 1, \gamma_h = 0$ ($\lambda_h = 1$), or $\tau_h = 1/f^2$ for the YaRN start (f: YaRN's attention factor).

PoSE. For RQ2, PoSE (Zhu et al., 2024) trains on 4k-token sequences split at a random point, with a random offset added to the second piece's position ids, so relative distances span up to L'. Unlike the original, which fine-tunes all weights and takes the second piece's text from later in the document, we train only ϕ and keep the text contiguous.

2.2 Data and metrics

Training and perplexity use PG-19 (Rae et al., 2020), a corpus of long English books (`emozilla/pg19` mirror; OLMo-2 tokenizer; non-overlapping chunks, tails dropped): 100 training chunks of 16k tokens (at most one per book; seed 0), and 50 validation and 50 test chunks at each of 4k / 8k / 16k. Retrieval is used only for evaluation, with two synthetic benchmarks. Passkey (Mohtashami and Jaggi, 2023) inserts a random five-digit key into repeated filler at a controlled depth; we score exact key recovery (750 test / 150 validation prompts; seeds 1 / 0). RULER (Hsieh et al., 2024): we evaluate the eight NIAH task variants at 4k / 8k / 16k (500 test / 100 validation samples per task and length; seeds 42 / 1234), with prompts from RULER's scripts and official string-match scoring under greedy decoding; inference uses our own Hugging Face code, since RULER's vLLM pipeline cannot run

our patched attention. We report PG-19 perplexity (4k also checks short-context regression), passkey accuracy, and the mean RULER NIAH score. We treat a method as successful if it improves long-context PPL and retrieval over strong training-free positional baselines while losing at most about one point of 4k PPL relative to the unmodified model.

2.3 Training details and compute

We train only ϕ with next-token cross-entropy on the 16k PG-19 chunks (batch size 1, 500 steps, AdamW (Loshchilov and Hutter, 2019), no weight decay, grad clip 1.0, bf16). We sweep LR $\in \{10^{-2}, 3\times10^{-2}\}$ and select it by validation PPL at 16k. Baselines are stock RoPE, PI (factor 4), default YaRN, and grid-tuned YaRN (factor 4, β_{fast}=64, β_{slow}=2, attention factor 1.2), chosen from 144 settings by validation perplexity and passkey accuracy, the latter on only six prompts at 16k, so it may not be optimal. A training run takes 12–15 minutes on one A100 / L40S-class GPU, the YaRN grid about 50 minutes, and full test evaluation, dominated by RULER, up to a few GPU-hours per checkpoint.

3 Results

All models extend the frozen OLMo-2-1B from 4k to 16k tokens. Table 1 gives test results for the baselines and three families of learned models: frequencies only, frequencies with a learned temperature (from $a = 0$ or from YaRN), and the latter trained with PoSE on 4k text only.

Learned frequencies need a learned temperature. From $a = 0$, frequencies alone beat tuned YaRN on 16k perplexity only per layer and per head; they are worse at 4k and fail at retrieval at 16k (Appendix, Figure 1). Adding a learned temperature, with everything else unchanged, restores 16k retrieval for the shared and per-layer models (Appendix, Figure 2). Every granularity then beats tuned YaRN on perplexity at all lengths, and shared and per-layer models also beat it on RULER at 16k. Per head from zero does not: with 16k parameters and 500 steps, it is likely under-trained.

The learned temperature is mostly a constant sharpening. From both starts, τ_h converges to similar values that sharpen attention by about the same factor at every length, while γ_h stays near zero for most heads, so attention grows only slightly sharper from 4k to 16k (Table 2). The temperature thus acts more like YaRN's fixed attention factor than like length-dependent scaling: it helps, but we cannot yet say it corrects flatter attention as such. A constant-τ run will test this. The median scale already exceeds 1 below L ($\lambda_{\leq 4k}$ in Table 2), which may account for part of the 4k cost.

Run	Start	$\lambda_{\leq 4k}$	Growth	Flat	RULER
Long, shared	a=0	1.18	1.02 (0.33–2.06)	1	36.8
Long, layer	a=0	1.16	1.04 (0.24–2.51)	1	33.3
Long, head	a=0	1.15	1.11 (0.06–2.04)	5	5.9
Long, shared	YaRN	1.15	1.02 (0.70–1.64)	0	36.2
Long, layer	YaRN	1.14	1.04 (0.70–1.46)	0	35.6
Long, head	YaRN	1.14	1.03 (0.77–1.37)	0	53.1
PoSE, shared	a=0	1.13	1.04 (0.10–2.42)	21	0.0
PoSE, layer	a=0	1.12	1.06 (0.09–3.30)	14	0.0
PoSE, head	a=0	1.15	1.06 (0.03–2.12)	8	0.1
PoSE, shared	YaRN	1.11	1.03 (0.17–2.13)	7	0.1
PoSE, layer	YaRN	1.11	1.07 (0.12–1.71)	6	0.0
PoSE, head	YaRN	1.11	1.05 (0.16–1.49)	5	1.1

Table 2: Learned temperatures over all 256 heads. $\lambda_{\leq 4k} = (\ln 2)^{\gamma_h}/\tau_h$: median query scale up to 4k (the YaRN start begins at 1.30, from zero at 1). Growth: $\lambda_h(16k)/\lambda_h(4k) = (\ln 5/\ln 2)^{\gamma_h}$, median (min–max); >1 is sharper at 16k. Flat: heads at least $2\times$ flatter at 16k. RULER: test average at 16k.

		Depth of key				
Model	Length	0	.25	.5	.75	1
Freq. only, layer	8k	100	100	98	100	100
	16k	0	0	0	0	0
YaRN, tuned	16k	88	90	76	52	96
+ temp., shared	16k	10	100	100	100	96

Table 3: Passkey accuracy (%) by key depth (0: start of input, 1: just before the question), 50 prompts per cell; models from a=0.

Why frequency-only models fail at 16k. Not because the key is too far away: at 16k they fail even when the key sits just before the question, while at 8k they succeed at every depth (Table 3). Instead of answering, they continue the filler text, with either prompt format we tried. One untested explanation is that some slow bands learned to rotate faster than in pretraining (Appendix, Figure 3), so distant tokens can look close.

YaRN is a good starting point. Starting from YaRN's frequencies and temperature beats tuned YaRN on 16k perplexity and RULER at every granularity, and beats the matching $a = 0$ model on 16k perplexity; on 16k RULER it wins by a wide margin per head and is on par for shared. Only models started from YaRN stay within one point of the original model's 4k perplexity, and they recover

much of the 4k RULER score that tuned YaRN loses.

PoSE works up to 8k but fails to retrieve at 16k. Trained only on 4k text, PoSE models beat tuned YaRN on perplexity and match long-data training on RULER up to 8k, but at 16k their retrieval collapses. Table 2 suggests why: they learn more heads with strongly negative γ_h, whose attention becomes flatter at 16k, which supports our hypothesis that temperatures are hard to learn from short text alone.

Perplexity does not predict retrieval. The frequency-only and PoSE failures above come with perplexity similar to or better than tuned YaRN (Appendix, Figure 4): perplexity mostly rewards using the last few sentences, while retrieval needs text far back. The two retrieval tests also disagree. The per-head model started from YaRN is best on RULER but fails passkey at 16k, while the shared models pass it as often as tuned YaRN. The haystack does not explain this: the same model does well on RULER's repeated-sentence haystack. We take RULER, which averages eight tasks with 500 samples each, as the main measure and leave the passkey gap to future analysis (§4).

Summary and caveats. Our best model, per head and started from YaRN, meets the success criterion on perplexity and RULER. The results suggest selecting models by retrieval rather than perplexity, and not learning the temperature from short text alone. Two attributions remain open: gains over training-free tuned YaRN credit training as much as the parameterization (an ablation learning only the temperature on YaRN frequencies is next), and the YaRN start's gain may come from its temperature initialization rather than its frequencies. Each configuration was run with one seed, many perplexity gaps are only 1–2 points, each test length uses few books, and learning rates were selected on validation perplexity only.

4 Future Work

Our remaining work follows the proposal's RQ2–RQ4 path and tightens evaluation.

Evaluation. We will broaden RULER beyond our current eight NIAH tasks and compare more closely to the official evaluation stack. We will also break results down per task (not only averages) and plot accuracy by needle depth/position for passkey and NIAH-style probes.

PoSE (RQ2). We will reimplement PoSE (Zhu et al., 2024) closer to the original recipe: short contiguous chunks with deleted middle text (or multiple gaps), so skipped position IDs correspond to skipped content. Our current variant only inserts a gap in contiguous text (pushes neighboring tokens apart without removing content); we will treat that as an ablation against true PoSE. As an extra idea, we will try random/uniform gap sizes and ask whether gap length should be chosen in an informed way (e.g., matching target relative distances) without defeating PoSE's goal of cheaper short-sequence training. We will log validation loss alongside training for these runs. To test whether negative γ_h causes the 16k failure, we will clamp $\gamma_h \geq 0$ at evaluation, which needs no retraining.

DaRoPE (RQ3). Learned frequencies still tie slow-band angles to position, so they grow without bound beyond L. Following DaRoPE (Lévy et al., 2026), each head will replace position m on the slow bands with a learned, content-dependent coordinate

$$c_h(x_m) = L\,\sigma\Big(w_h^\top \tilde{x}_m + \alpha_h \operatorname{logit}\tfrac{m+0.5}{L}\Big), \quad (4)$$

where $\tilde{x}_m$ is the RMS-normalized hidden state. The coordinate stays in $[0, L]$, so angles never leave the pretraining range, and content can bring related tokens close regardless of distance. At $w_h = 0$, $\alpha_h = 1$, $c_h(x_m) = m + 0.5$ for $m < L$, so training starts from the original model (∼525k parameters). DaRoPE has only been trained from scratch; we test retrofitting it onto a frozen model.

Analysis. To explain retrieval wins/losses, we will compare attention weights on needle tokens for strong vs. weak configurations.

RQ4 (stretch). If learned positions still fall short, is the rest of the problem in the backbone weights? To find out, we will unfreeze them step by step, first with LoRA (Hu et al., 2022) on the query and key projections, then with full fine-tuning, and measure how much each step adds.

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A Additional results

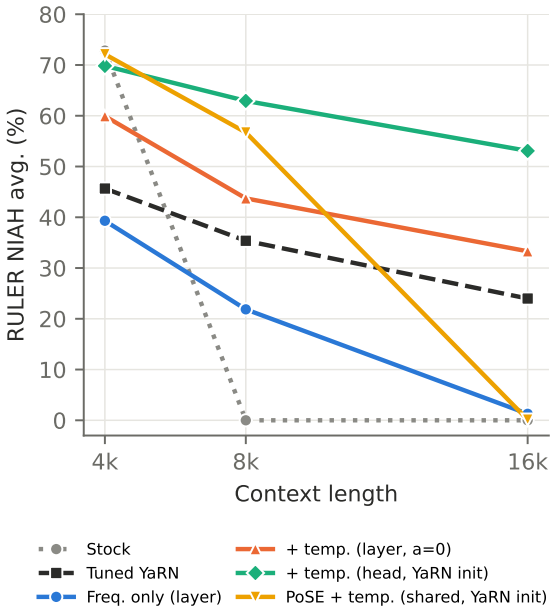


Figure 1: RULER score vs. input length for a representative model of each family. Frequencies only (blue) and PoSE (yellow) fall to zero at 16k; the models with a learned temperature shown here stay above tuned YaRN (dashed).

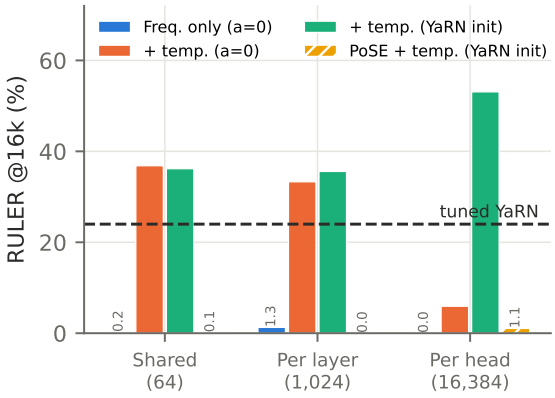


Figure 2: RULER at 16k per granularity. Blue vs. orange differ only in the learned temperature; green starts both at YaRN; hatched yellow is PoSE (trained on 4k text). Dashed: tuned YaRN.

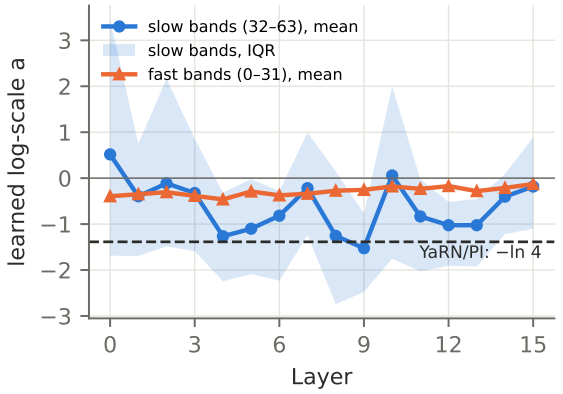


Figure 3: Learned log frequency scales a of the per-layer, frequencies-only model ($a < 0$: slower rotation, $a = 0$: unchanged). Line: mean over bands; shaded: middle 50% of slow bands. Dashed: the single value YaRN and PI use for slow bands.

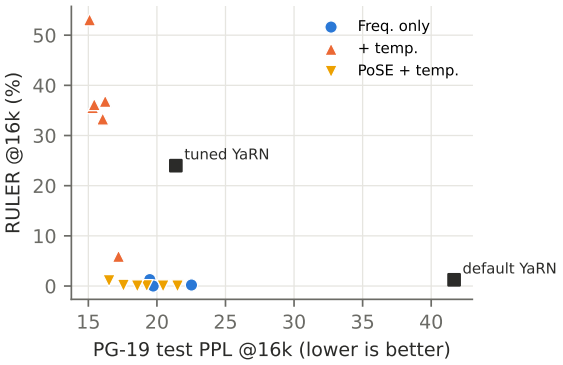


Figure 4: PG-19 test perplexity vs. RULER score at 16k. Each marker is one trained model: frequencies only (blue), + temperature (orange), PoSE + temperature (yellow). The original model and PI are off the scale and not shown.